Exercises: Multivariate Taylor Expansion, Critical Points & Extrema – Answer Key

Mathematical Foundations for Machine Learning

Taylor Expansion — Computation

Exercise 1.1 [Easy]
Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined as $f(x, y) = e^{x+y}$. Compute the second-order Taylor expansion of f around $x_0 = (0, 0)$.

Solution

We compute $f(0, 0) = 1$. The gradient is $\nabla f = (e^{x+y}, e^{x+y})^T$, so $\nabla f(0, 0) = (1, 1)^T$.
The Hessian is:

$$Hf(x, y) = \begin{pmatrix} e^{x+y} & e^{x+y} \\ e^{x+y} & e^{x+y} \end{pmatrix}, \quad Hf(0, 0) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Plugging into the second-order Taylor formula $f(x) \approx f(x_0) + \nabla f(x_0)^T h + \frac{1}{2} h^T Hf(x_0) h$ with $h = (x, y)^T$:

$$f(x, y) \approx 1 + x + y + \frac{1}{2}(x^2 + 2xy + y^2) = 1 + x + y + \frac{1}{2}(x + y)^2.$$

Exercise 1.2 [Easy]
Let $f(x, y) = \sin(x)\cos(y)$. Compute the second-order Taylor expansion around $(0, 0)$.

Solution

We have $f(0, 0) = 0$. The gradient:

$$\nabla f = \begin{pmatrix} \cos(x)\cos(y) \\ -\sin(x)\sin(y) \end{pmatrix}, \quad \nabla f(0, 0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The Hessian:

$$Hf = \begin{pmatrix} -\sin(x)\cos(y) & -\cos(x)\sin(y) \\ -\cos(x)\sin(y) & -\sin(x)\cos(y) \end{pmatrix}, \quad Hf(0, 0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Thus the second-order expansion is simply $f(x, y) \approx x$. The quadratic term vanishes, so higher-order terms would be needed for a more accurate approximation.

Exercise 1.3 [Medium]
Given $f : \mathbb{R}^n \to \mathbb{R}$ defined by $f(x) = \sqrt{1 + \|x\|^2}$. Compute the second-order Taylor expansion at $x_0 = 0$.

(Exam 2026, Exercise 2b)

Solution

Step 1: $f(0) = \sqrt{1+0} = 1$.
Step 2: We compute $\frac{\partial f}{\partial x_i} = \frac{x_i}{\sqrt{1+\|x\|^2}}$, so $\nabla f(0) = 0$.
Step 3: The Hessian entries are:

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\delta_{ij}}{\sqrt{1+\|x\|^2}} - \frac{x_i x_j}{(1+\|x\|^2)^{3/2}}.$$

At $x = 0$: $Hf(0) = I_n$ (the identity matrix).
Step 4: Taylor expansion:

$$f(x) \approx 1 + 0 + \frac{1}{2} x^T I_n x = 1 + \frac{1}{2}\|x\|^2.$$

Exercise 1.4 [Medium]
Let $f : \mathbb{R}^2 \to \mathbb{R}$ with $f(x, y) = \log(1+x^2+y^2+xy)$. Compute the second-order Taylor expansion around $(0, 0)$ and verify your result by comparing with the expansion of $\log(1 + u)$ for small u.

Solution

Step 1: $f(0, 0) = \log(1) = 0$.
Step 2: Let $g = 1 + x^2 + y^2 + xy$. Then $\frac{\partial f}{\partial x} = \frac{2x+y}{g}$ and $\frac{\partial f}{\partial y} = \frac{2y+x}{g}$. At $(0, 0)$: $\nabla f(0, 0) = (0, 0)^T$.
Step 3: Using the quotient rule:

$$\frac{\partial^2 f}{\partial x^2} = \frac{2g - (2x+y)^2}{g^2}, \quad \frac{\partial^2 f}{\partial y^2} = \frac{2g - (2y+x)^2}{g^2}, \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{g - (2x+y)(2y+x)}{g^2}.$$

At $(0, 0)$: $Hf(0, 0) = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.
Step 4: $f(x, y) \approx 0 + 0 + \frac{1}{2}(2x^2 + 2xy + 2y^2) = x^2 + xy + y^2$.
Verification: Setting $u = x^2 + y^2 + xy$, we get $\log(1 + u) \approx u = x^2 + y^2 + xy$ for small u, which matches. ✓

Exercise 1.5 [Hard]
Let $A \in \mathbb{R}^{n\times n}$ be symmetric and positive definite. Define $f : \mathbb{R}^n \to \mathbb{R}$ by $f(x) = \log(x^T Ax)$. Compute the second-order Taylor expansion around any point x_0 with $x_0^T Ax_0 \neq 0$.

Solution

Let $q_0 = x_0^T A x_0 > 0$.
Step 1: $f(x_0) = \log(q_0)$.
Step 2: We use $\nabla_x(x^T A x) = 2Ax$ (since A symmetric). By the chain rule:

$$\nabla f(x) = \frac{2Ax}{x^T A x}, \quad \nabla f(x_0) = \frac{2Ax_0}{q_0}.$$

Step 3: For the Hessian, differentiate $\nabla f(x) = 2Ax \cdot (x^T A x)^{-1}$:

$$Hf(x) = \frac{2A}{x^T A x} - \frac{4Axx^T A}{(x^T A x)^2}.$$

At x_0: $Hf(x_0) = \frac{2A}{q_0} - \frac{4Ax_0x_0^T A}{q_0^2}$.
Step 4: With $h = x - x_0$:

$$f(x) \approx \log(q_0) + \frac{2(Ax_0)^T h}{q_0} + \frac{1}{2}h^T \left(\frac{2A}{q_0} - \frac{4Ax_0x_0^T A}{q_0^2}\right) h.$$

Critical Points — Finding and Classifying

Exercise 2.1 [Easy]
Find all critical points of $f(x, y) = x^2 + y^2 - 2x - 4y + 5$ and classify them as local min, local max, or saddle point.

Solution

Step 1 (Find critical points):

$$\nabla f = \begin{pmatrix} 2x - 2 \\ 2y - 4 \end{pmatrix} = 0 \implies (x_0, y_0) = (1, 2).$$

Step 2 (Hessian):

$$Hf = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2I.$$

Since $Hf \succ 0$ (both eigenvalues are $2 > 0$), the point $(1, 2)$ is a **strict local minimum** with $f(1, 2) = 0$.

Exercise 2.2 [Easy]
Find and classify all critical points of $f(x, y) = x^3 - 3xy + y^3$.

Solution

Step 1:

$$\nabla f = \begin{pmatrix} 3x^2 - 3y \\ -3x + 3y^2 \end{pmatrix} = 0 \implies y = x^2 \text{ and } x = y^2.$$

Substituting: $x = (x^2)^2 = x^4$, so $x^4 - x = 0$, giving $x(x^3 - 1) = 0$. Critical points: $(0,0)$ and $(1,1)$.

Step 2:

$$Hf = \begin{pmatrix} 6x & -3 \\ -3 & 6y \end{pmatrix}.$$

At $(0,0)$: $Hf = \begin{pmatrix} 0 & -3 \\ -3 & 0 \end{pmatrix}$ with eigenvalues $\pm 3 \Rightarrow$ indefinite $\Rightarrow$ **saddle point**.

At $(1,1)$: $Hf = \begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}$ with eigenvalues $3, 9 \Rightarrow$ positive definite $\Rightarrow$ **strict local minimum**.

Exercise 2.3 [Medium]

Find and classify all critical points of $f(x,y) = (x^2 + y^2)e^{-(x^2+y^2)}$.

Solution

Let $r^2 = x^2 + y^2$, so $f = r^2 e^{-r^2}$.

Step 1: $\frac{\partial f}{\partial x} = 2xe^{-r^2} - 2xr^2e^{-r^2} = 2xe^{-r^2}(1 - r^2)$, and similarly for y. So $\nabla f = 2e^{-r^2}(1 - r^2)\begin{pmatrix} x \\ y \end{pmatrix} = 0$.

This gives: $(x,y) = (0,0)$ or $r^2 = x^2 + y^2 = 1$.

Step 2 (Classification at origin): Computing the Hessian at $(0,0)$:

$$Hf(0,0) = 2I \succ 0 \implies \textbf{strict local minimum } (f = 0).$$

Step 3 (Circle $r = 1$): Along the circle, $f = e^{-1}$ is constant. The Hessian restricted to the radial direction gives $\frac{\partial^2}{\partial r^2}(r^2e^{-r^2})\big|_{r=1} = e^{-1}(2 - 8 + 4) = -2e^{-1} < 0$. So these are **local maxima** with value e^{-1}.

Note: the function tends to 0 as $\|x\| \to \infty$ and is nonneg everywhere, so the origin is also the global minimum.

Exercise 2.4 [Medium]

Consider the function from Exam 2021 Exercise 2a in a related form: let $f(x,y) = x^2 + xy + y^2 - 3x - 3y$. Find all critical points, classify them, and compute the value of f at each.

Then compute the directional derivative at the critical point in the direction $v = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and explain why the result makes sense.

Solution

Step 1 (Critical points):

$$\nabla f = \begin{pmatrix} 2x + y - 3 \\ x + 2y - 3 \end{pmatrix} = 0 \implies 2x + y = 3 \text{ and } x + 2y = 3.$$

Solving: $x = y = 1$, so the unique critical point is $(1, 1)$ with $f(1, 1) = 1+1+1-3-3 = -3$.
Step 2 (Classification):

$$Hf = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Eigenvalues: $\lambda = 1, 3$ (both positive) $\Rightarrow Hf \succ 0 \Rightarrow$ **strict local minimum**.
Step 3 (Directional derivative): $D_v f(1, 1) = \nabla f(1, 1)^T v = (0, 0)^T v = 0$.
This is expected: at a critical point $\nabla f = 0$, so the directional derivative vanishes in *every* direction.

Exercise 2.5 [Hard]
Consider $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x, y) = x^2 + y^2(1 - x)^3$.

(a) Find all critical points and classify them using the Hessian.

(b) Show that f has no global minimum by finding a direction along which $f \to -\infty$.

(c) Does this contradict the result that, for $f \in C^1(\mathbb{R})$, a unique critical point that is a local minimum must be the global minimum? Explain carefully.

Solution

(a) $\nabla f = \begin{pmatrix} 2x - 3y^2(1 - x)^2 \\ 2y(1 - x)^3 \end{pmatrix} = 0.$
From the second equation: $y = 0$ or $x = 1$. If $x = 1$: the first equation gives $2 = 0$, contradiction. If $y = 0$: the first equation gives $2x = 0$, so $x = 0$. Unique critical point: $(0, 0)$.
$Hf(0, 0) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \succ 0$, so $(0, 0)$ is a **strict local minimum**.
(b) Fix $y = c \neq 0$ and let $x \to +\infty$: $f(x, c) = x^2 + c^2(1 - x)^3$. The $(1 - x)^3$ term dominates (cubic vs. quadratic), so $f(x, c) \to -\infty$. Thus no global minimum exists.
(c) No contradiction. The theorem applies to $f : \mathbb{R} \to \mathbb{R}$ only (single variable). For $f : \mathbb{R}^n \to \mathbb{R}$ with $n \geq 2$, a unique local minimum that is also the unique critical point need *not* be a global minimum. The one-dimensional proof uses the intermediate value theorem and Rolle's theorem on intervals, which do not generalize directly to higher dimensions.

Local vs. Global Extrema

Exercise 3.1 [Easy]
State the necessary condition and the sufficient condition for a critical point x_0 of $f \in C^2(\Omega)$ to be a local minimum. Give a concrete example where the necessary condition is satisfied but the sufficient condition is not.

Solution

Necessary condition: If x_0 is a local minimum, then $\nabla f(x_0) = 0$ and $Hf(x_0) \succeq 0$ (positive semi-definite).
Sufficient condition: If $\nabla f(x_0) = 0$ and $Hf(x_0) \succ 0$ (positive definite), then x_0 is a strict local minimum.
Counterexample: $f(x) = x^4$ in $\mathbb{R}$. At $x_0 = 0$: $f'(0) = 0$, $f''(0) = 0$ (PSD but not PD). Here $x_0 = 0$ is a local (and global) minimum, but the sufficient condition does not apply. To see a case where PSD does *not* imply local minimum, consider $f(x, y) = x^2 - y^4$ at the origin: $Hf(0, 0) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \succeq 0$ but $(0, 0)$ is a saddle point.

Exercise 3.2 [Medium]
Let $f \in C^1(\mathbb{R})$ have a single critical point x_0 which is a local minimum. Prove that x_0 is the global minimum.

Hint: Use the theorem of Rolle by contradiction.
(Exam 2021 style — proof question)

Solution

Proof by contradiction. Suppose x_0 is not the global minimum, i.e., there exists x^* with $f(x^*) < f(x_0)$. WLOG assume $x^* < x_0$.
Since x_0 is a local minimum, there exists $\varepsilon > 0$ such that $f(x) \geq f(x_0)$ for $x \in (x_0 - \varepsilon, x_0 + \varepsilon)$.
Define z as the smallest number in $(x^*, x_0]$ with $f(z) = f(x_0)$. Such z exists by the intermediate value theorem (since $f(x^*) < f(x_0)$ and f is continuous).
Note $z < x_0$ because if $z = x_0$, then for points slightly to the left of x_0 we'd have $f < f(x_0)$, contradicting local minimality.
By Rolle's theorem, there exists $w \in (z, x_0)$ with $f'(w) = 0$. Since $w \neq x_0$, this gives a second critical point, contradicting the uniqueness assumption. $\square$

Exercise 3.3 [Medium]
Prove that any local minimum of a convex function $f : \Omega \to \mathbb{R}$ over a convex set Ω is also a global minimum.

Hint: Prove by contradiction using the definition of convexity.
(Exam 2021, Exercise 3a)

Solution

Proof by contradiction. Suppose x is a local minimum but not a global minimum. Then there exists $y \in \Omega$ with $f(y) < f(x)$.
Since Ω is convex, for any $\lambda \in (0, 1)$, the point $z_\lambda = \lambda y + (1 - \lambda)x \in \Omega$. By convexity of f:

$$f(z_\lambda) \leq \lambda f(y) + (1 - \lambda)f(x) < \lambda f(x) + (1 - \lambda)f(x) = f(x).$$

Now, $\|z_\lambda - x\| = \lambda\|y - x\|$. For any neighborhood radius $r > 0$, choosing $\lambda < \frac{r}{\|y-x\|}$ ensures $z_\lambda \in B(x, r)$.
Thus we have found a point z_λ in every neighborhood of x with $f(z_\lambda) < f(x)$, contradicting the assumption that x is a local minimum. $\square$

Exercise 3.4 [Hard]

Let $f \in C^1(\mathbb{R}^n)$ have a unique critical point x_0 which is a local minimum. Suppose additionally that there exists $r > 0$ such that $f(x) > f(x_0)$ for all $x \in \mathbb{R}^n \setminus B(x_0, r)$. Prove that x_0 is the global minimum.

Hint: Use compactness of the closed ball and the extreme value theorem.

Solution

Proof. The closed ball $\overline{B}(x_0, r)$ is closed and bounded, hence compact. Since f is continuous, by the extreme value theorem f attains its minimum on $\overline{B}(x_0, r)$ at some point x^*.
We claim x^* is in the interior of the ball. For any $x \in \partial B(x_0, r)$ (the boundary), we have $f(x) > f(x_0)$ by hypothesis. Therefore x^* cannot be on the boundary, so $x^* \in B(x_0, r)$ (the open ball).
Since x^* is a minimum of f on $\overline{B}(x_0, r)$ and lies in the interior, x^* must satisfy $\nabla f(x^*) = 0$. But x_0 is the *unique* critical point, so $x^* = x_0$.
Thus $f(x_0) \leq f(x)$ for all $x \in \overline{B}(x_0, r)$. Combined with $f(x) > f(x_0)$ for $x \notin B(x_0, r)$, we get $f(x_0) \leq f(x)$ for all $x \in \mathbb{R}^n$. $\square$

Positive Semi-Definite Hessians and Degenerate Cases

Exercise 4.1 [Easy]
Classify the critical point at the origin for each of the following functions:

(a) $f(x, y) = x^2 + y^4$

(b) $f(x, y) = x^2 - y^4$

(c) $f(x, y) = x^4 + y^4$

Note that the Hessian is only semi-definite at the origin for each. Can the Hessian test conclude anything? Determine the nature of each critical point from the definition.

Solution

For all three, $\nabla f(0, 0) = 0$. The Hessians at $(0, 0)$:
(a) $Hf = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \succeq 0$ (PSD). The Hessian test is inconclusive. But $f(x, y) = x^2 + y^4 \geq 0 = f(0, 0)$, so $(0, 0)$ is a **global minimum** by definition.
(b) $Hf = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \succeq 0$ (PSD). Inconclusive. But $f(0, y) = -y^4 < 0$ for $y \neq 0$ and $f(x, 0) = x^2 > 0$ for $x \neq 0$. So $(0, 0)$ is a **saddle point**.
(c) $Hf = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \succeq 0$ (PSD). Inconclusive. But $f(x, y) = x^4 + y^4 \geq 0 = f(0, 0)$, so $(0, 0)$ is a **global minimum**.
Lesson: A PSD Hessian at a critical point is a necessary condition for a local min, but tells us nothing definitive. One must argue from the definition or higher-order terms.

Exercise 4.2 [Medium]
Let $\|u\| = 1$ and define $f : \mathbb{R}^n \to \mathbb{R}$ by

$$f(x) = \langle x, (I - uu^T)x \rangle - \langle u, x \rangle^4.$$

Show that $\nabla f(0) = 0$ and $Hf(0) \succeq 0$, but $x_0 = 0$ is **not** a local minimum.
(This is the counterexample from the lecture notes.)

Solution

Gradient: $\nabla f(x) = 2(I - uu^T)x - 4\langle u, x\rangle^3 u$. At $x = 0$: $\nabla f(0) = 0$. ✓
Hessian: $Hf(x) = 2(I - uu^T) - 12\langle u, x\rangle^2 uu^T$. At $x = 0$: $Hf(0) = 2(I - uu^T)$.
Check PSD: $\langle h, Hf(0)h\rangle = 2\|h\|^2 - 2\langle u, h\rangle^2 \geq 0$ by Cauchy-Schwarz (since $\|u\| = 1$). Equality holds for $h = \alpha u$. So $Hf(0) \succeq 0$. ✓
Not a local min: For any $\varepsilon > 0$, take $x = \varepsilon u$:

$$f(\varepsilon u) = \varepsilon^2 \underbrace{\langle u, (I - uu^T)u\rangle}_{=1-1=0} - \varepsilon^4 = -\varepsilon^4 < 0 = f(0).$$

So in every neighborhood of 0, there exist points with $f < f(0)$. Thus 0 is not a local minimum (it is in fact a saddle point, since $f(\varepsilon v) > 0$ for $v \perp u$). □

Exercise 4.3 [Hard]
Let $f \in C^2(\mathbb{R}^n)$ with $\nabla f(x_0) = 0$. Prove the following:

(a) (Necessary condition) If x_0 is a local minimum, then $Hf(x_0) \succeq 0$.

(b) (Sufficient condition) If $Hf(x_0) \succ 0$, then x_0 is a strict local minimum.

Use the second-order Taylor expansion with remainder.

Solution

(a) Necessary condition. Since x_0 is a local min, there exists $\varepsilon > 0$ with $f(x_0+h) \geq f(x_0)$ for all $\|h\| < \varepsilon$. By Taylor:

$$0 \leq f(x_0 + \alpha h) - f(x_0) = \alpha \underbrace{\langle \nabla f(x_0), h\rangle}_{=0} + \frac{\alpha^2}{2}\langle h, Hf(x_0 + \theta\alpha h)\, h\rangle$$

for some $\theta \in (0, 1)$. Dividing by $\alpha^2/2 > 0$: $\langle h, Hf(x_0 + \theta\alpha h)\, h\rangle \geq 0$.
Taking $\alpha \to 0$ and using continuity of Hf:

$$\langle h, Hf(x_0)\, h\rangle = \lim_{\alpha\to 0}\langle h, Hf(x_0 + \theta\alpha h)\, h\rangle \geq 0.$$

Since this holds for all h, $Hf(x_0) \succeq 0$. □
(b) Sufficient condition. Since $Hf(x_0) \succ 0$, we have $\langle h, Hf(x_0)h\rangle > 0$ for all $h \neq 0$. By continuity of Hf, there exists $\varepsilon > 0$ such that $\langle h, Hf(x)h\rangle > 0$ for all $h \neq 0$ and $x \in B(x_0, \varepsilon)$.
For any $h \neq 0$ with $\|h\| < \varepsilon$, by Taylor there exists $\theta \in (0, 1)$:

$$f(x_0 + h) - f(x_0) = \underbrace{\langle \nabla f(x_0), h\rangle}_{=0} + \frac{1}{2}\underbrace{\langle h, Hf(x_0 + \theta h)\, h\rangle}_{>0} > 0.$$

Since $\|\theta h\| < \varepsilon$, the point $x_0 + \theta h \in B(x_0, \varepsilon)$, so the quadratic form is positive. Thus $f(x_0 + h) > f(x_0)$ for all $0 < \|h\| < \varepsilon$, making x_0 a strict local minimum. □

Mixed Challenging Exercises

Exercise 5.1 **[Hard]**
Let $W \in \mathbb{R}^{m\times n}$ and define the loss function of a single-layer linear network:

$$L(W) = \|y - Wx\|_2^2$$

where $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$ are fixed.

(a) Compute $\nabla_W L$ and find all critical points.

(b) Compute the Hessian. *Hint:* treat W as a vector in $\mathbb{R}^{mn}$ via $\mathrm{vec}(W)$, and use the identity $\mathrm{vec}(ABC) = (C^T \otimes A)\mathrm{vec}(B)$.

(c) Classify the critical point(s). Under what condition on x is the critical point a strict local minimum?

Solution

(a) Expanding: $L(W) = (y - Wx)^T(y - Wx) = y^Ty - 2y^TWx + x^TW^TWx$.
Using matrix calculus: $\nabla_W L = -2(y - Wx)x^T = -2rx^T$ where $r = y - Wx$.
Setting $\nabla_W L = 0$: $(y - Wx)x^T = 0$. If $x \neq 0$, this requires $y - Wx = \alpha v$ for some $v \perp x$, but more precisely $(y - Wx) \in \ker(x^T) = \{0\}$ when we need it to hold as a matrix equation.
The critical points satisfy $Wxx^T = yx^T$, i.e., $W(xx^T) = yx^T$.
If $x \neq 0$: $W^* = yx^T(xx^T)^{-1} = \frac{yx^T}{\|x\|^2}$ is the unique solution restricted to $\mathrm{span}(x)$ (with freedom in $x^\perp$ directions).
(b) Using vectorization: $\mathrm{vec}(\nabla_W L) = -2(x \otimes I_m)\mathrm{vec}(y) + 2(xx^T \otimes I_m)\mathrm{vec}(W)$. The Hessian w.r.t. $\mathrm{vec}(W)$ is $H = 2(xx^T \otimes I_m)$.
(c) Eigenvalues of $xx^T \otimes I_m$: these are $\|x\|^2$ (with multiplicity m) and 0 (with multiplicity $m(n-1)$). So $H \succeq 0$ always, and $H \succ 0$ if and only if $n = 1$ and $x \neq 0$. For $n > 1$, the Hessian is PSD but not PD, reflecting the flat directions in W corresponding to $x^\perp$. The problem is convex, so the critical point is a global minimum, but not strict when $n > 1$.

Exercise 5.2 **[Hard]**
Let $A \in \mathbb{R}^{n\times n}$ be symmetric positive definite. Consider the function $f : \mathbb{R}^n \to \mathbb{R}$:

$$f(x) = \frac{1}{2}x^TAx - b^Tx$$

where $b \in \mathbb{R}^n$ is fixed.

(a) Find the unique critical point and prove it is a strict global minimum.

(b) Compute the second-order Taylor expansion of f around the minimizer x^*.

(c) Prove that $f(x) - f(x^*) = \frac{1}{2}(x - x^*)^TA(x - x^*)$ for all x. Interpret this geometrically in terms of the level sets of f.

Solution

(a) $\nabla f(x) = Ax - b = 0 \implies x^* = A^{-1}b$ (unique since A invertible).
$Hf(x) = A \succ 0$ for all x, so x^* is a strict local minimum. Since f is strongly convex ($A \succ 0$ implies f is $\lambda_{\min}(A)$-strongly convex), the local minimum is the unique global minimum.
(b) Since $\nabla f(x^*) = 0$ and $Hf = A$ (constant), the Taylor expansion is exact:

$$f(x) = f(x^*) + \frac{1}{2}(x - x^*)^T A(x - x^*).$$

There is no remainder since f is quadratic.
(c) Direct computation: $f(x) = \frac{1}{2}x^T Ax - b^T x$ and $f(x^*) = \frac{1}{2}(A^{-1}b)^T A(A^{-1}b) - b^T A^{-1}b = -\frac{1}{2}b^T A^{-1}b$.
Thus $f(x) - f(x^*) = \frac{1}{2}x^T Ax - b^T x + \frac{1}{2}b^T A^{-1}b$. Setting $h = x - x^* = x - A^{-1}b$:

$$\frac{1}{2}h^T Ah = \frac{1}{2}(x - A^{-1}b)^T A(x - A^{-1}b) = \frac{1}{2}x^T Ax - b^T x + \frac{1}{2}b^T A^{-1}b. \checkmark$$

Geometric interpretation: The level sets $\{x : f(x) = c\}$ are ellipsoids centered at x^*, with axes aligned with the eigenvectors of A and semi-axis lengths proportional to $1/\sqrt{\lambda_i}$. The condition number $\kappa(A) = \lambda_{\max}/\lambda_{\min}$ determines how elongated the ellipsoids are.

Exercise 5.3 [Hard]
Let $f \in C^2(\mathbb{R}^n)$ and suppose x_0 is a critical point with $Hf(x_0)$ having eigenvalues $\lambda_1 < 0 < \lambda_2$, with corresponding eigenvectors u_1, u_2. Using the Taylor expansion:

(a) Prove that x_0 is a saddle point by showing that f increases along one direction and decreases along another.

(b) In the context of training a neural network with loss function $L(\theta)$, explain why a saddle point might be more problematic than a local minimum for gradient descent. How does the second-order information from Hf help?

Solution

(a) Since $\nabla f(x_0) = 0$, Taylor gives for $h \neq 0$ small:

$$f(x_0 + \varepsilon h) - f(x_0) = \frac{\varepsilon^2}{2}\langle h, Hf(x_0 + \theta\varepsilon h)\, h\rangle.$$

By continuity of Hf, for sufficiently small ε:
Direction u_1 (negative eigenvalue):

$$f(x_0 + \varepsilon u_1) - f(x_0) = \frac{\varepsilon^2}{2}\langle u_1, Hf(x_0 + \theta\varepsilon u_1)u_1\rangle \approx \frac{\varepsilon^2}{2}\lambda_1 < 0.$$

More precisely, since $\langle u_1, Hf(x_0)u_1\rangle = \lambda_1 < 0$ and Hf is continuous, there exists $\delta > 0$ such that $\langle u_1, Hf(x)u_1\rangle < 0$ for $x \in B(x_0, \delta)$, so f strictly decreases along u_1.
Direction u_2 (positive eigenvalue): Similarly, $f(x_0 + \varepsilon u_2) - f(x_0) \approx \frac{\varepsilon^2}{2}\lambda_2 > 0$.
Since f decreases in one direction and increases in another, x_0 is neither a local min nor max, hence a **saddle point**. $\square$
(b) At a saddle point, $\nabla L = 0$ so gradient descent stalls. In high-dimensional loss landscapes, saddle points are exponentially more common than local minima (if the Hessian has n eigenvalues, the probability that all are positive decreases exponentially with n).
Second-order methods (Newton's method, natural gradient) use Hf to detect and escape saddle points: the negative eigenvalue direction u_1 indicates a direction of descent. Practically, techniques like adding noise to SGD or using negative curvature directions help escape saddle points. The eigenvalue magnitudes $|\lambda_i|$ quantify how "steep" the saddle is in each direction.